Rakuten Product Classification — Visualizations 3+

This notebook contains the **visualizations from the newer audit notebook starting with Figure 3**, but adapted so it can run inside the existing `liora-capstone` GitHub project.

The data is loaded at the beginning using the project's existing helper:

```
df = load_merged_data()
```

Figures 1 and 2 from the newer notebook are intentionally omitted.

1. Imports

```
Environment loaded successfully.
```

2. Load Data

```
Dataset shape: (84916, 5)
```

	designation	description	productid	imageid	prdtypecode
0	Olivia: Personalisiertes Notizbuch / 150 Seiten / Punktraster / Ca Din A5 / Rosen-Design	NaN	3804725264	1263597046	10
1	Journal Des Arts (Le) N° 133 Du 28/09/2001 - L'art Et Son Marche Salon D'art Asiatique A Paris - Jacques Barrere - F...	NaN	436067568	1008141237	2280
2	Grand Stylet Ergonomique Bleu Gamepad Nintendo Wii U - Speedlink Pilot Style	PILOT STYLE Touch Pen de marque Speedlink est 1 stylet ergonomique pour GamePad Nintendo Wii U. Pour un confort ...	201115110	938777978	50
3	Peluche Donald - Europe - Disneyland 2000 (Marionnette À Doigt)	NaN	50418756	457047496	1280
4	La Guerre Des Tuques	Luc a des idées de grandeur. Il veut organiser un jeu de guerre de boules de neige et s'arranger pour en &eci...	278535884	1077757786	2705

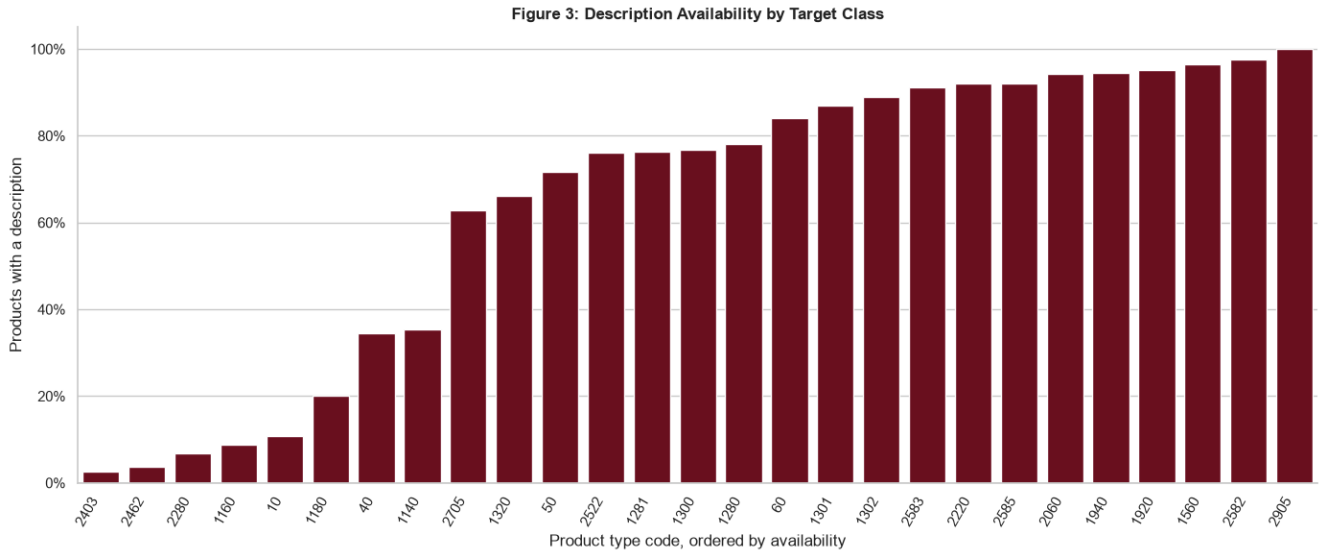
3. Audit Feature Construction

The following analytical features are required by Figures 3 onward.
The original dataframe `df` remains unchanged.

	designation	has_description	designation_chars	description_chars	total_text_chars
0	Olivia: Personalisiertes Notizbuch / 150 Seiten / Punktraster / Ca Din A5 / Rosen-Design	False	88	0	88
1	Journal Des Arts (Le) N° 133 Du 28/09/2001 - L'art Et Son Marche Salon D'art Asiatique A Paris - Jacques Barrere - F...	False	206	0	206
2	Grand Stylet Ergonomique Bleu Gamepad Nintendo Wii U - Speedlink Pilot Style	True	76	760	836
3	Peluche Donald - Europe - Disneyland 2000 (Marionnette À Doigt)	False	63	0	63
4	La Guerre Des Tuques	True	20	213	233

4. Figure 3 — Description Availability by Class

If missing descriptions are associated with the target, description availability itself may contain category-related information.



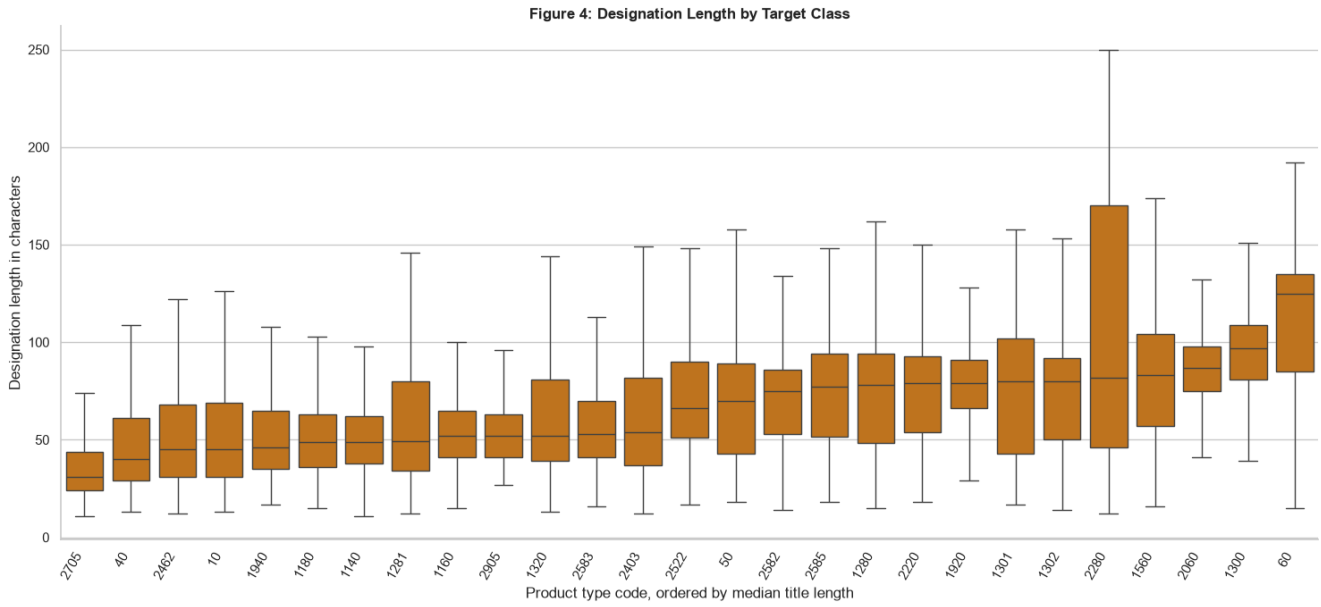
Finding 3 — missing descriptions are category-dependent

Observation. Class-level description availability ranges from 2.6% to 100.0%.

Quantitative evidence. χ^2=43,757.8, p=0, Cramér's V=0.718.

Business implication. Description presence may improve classification, but it can also reflect seller behaviour rather than product semantics.

5. Figure 4 — Designation Length by Class



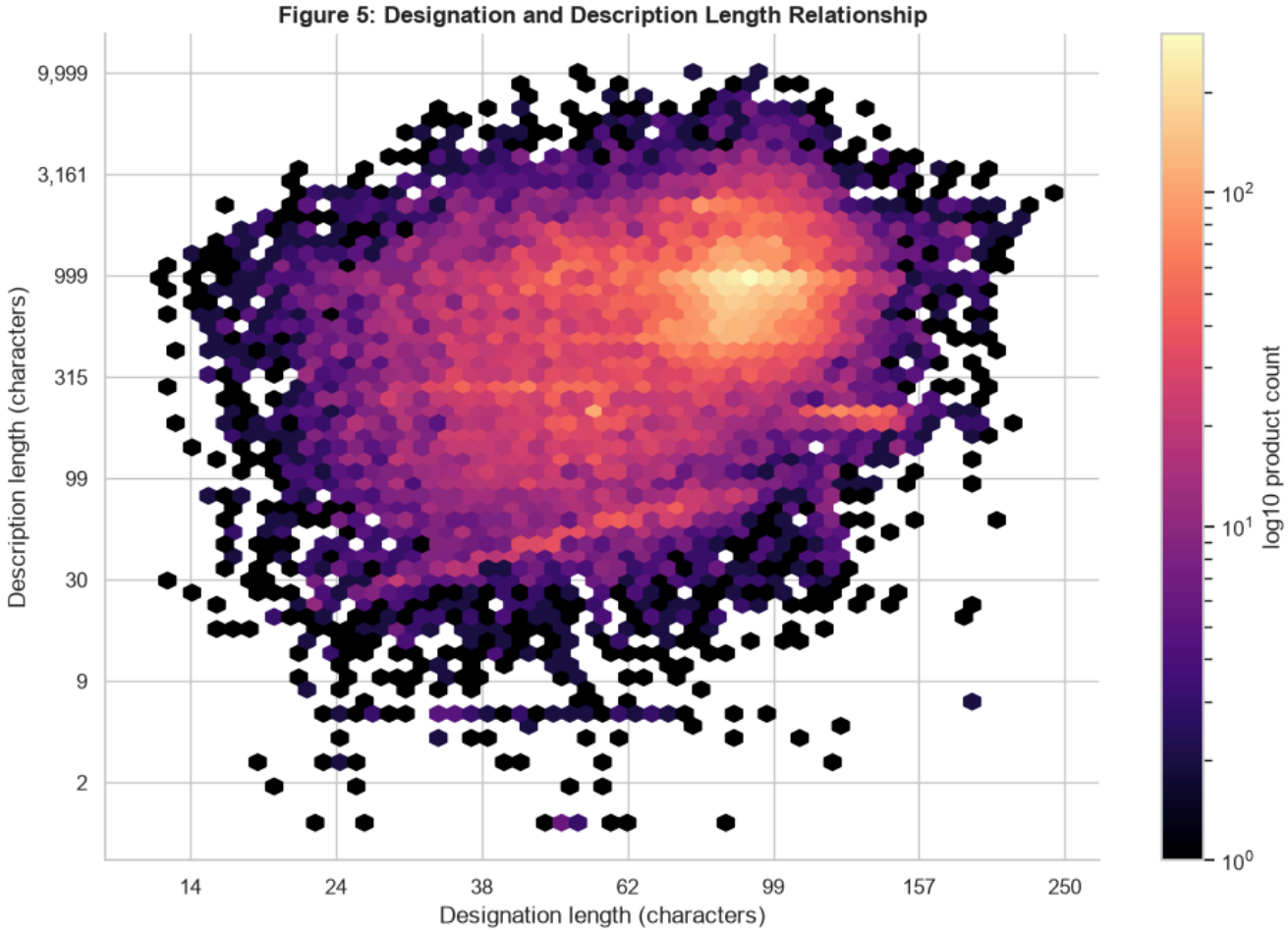
Finding 4 — title length contains target-related structure

Observation. Median designation length by class ranges from 31 to 125 characters.

Quantitative evidence. Kruskal–Wallis H=19,025.3, p=0; epsilon-squared=0.224.

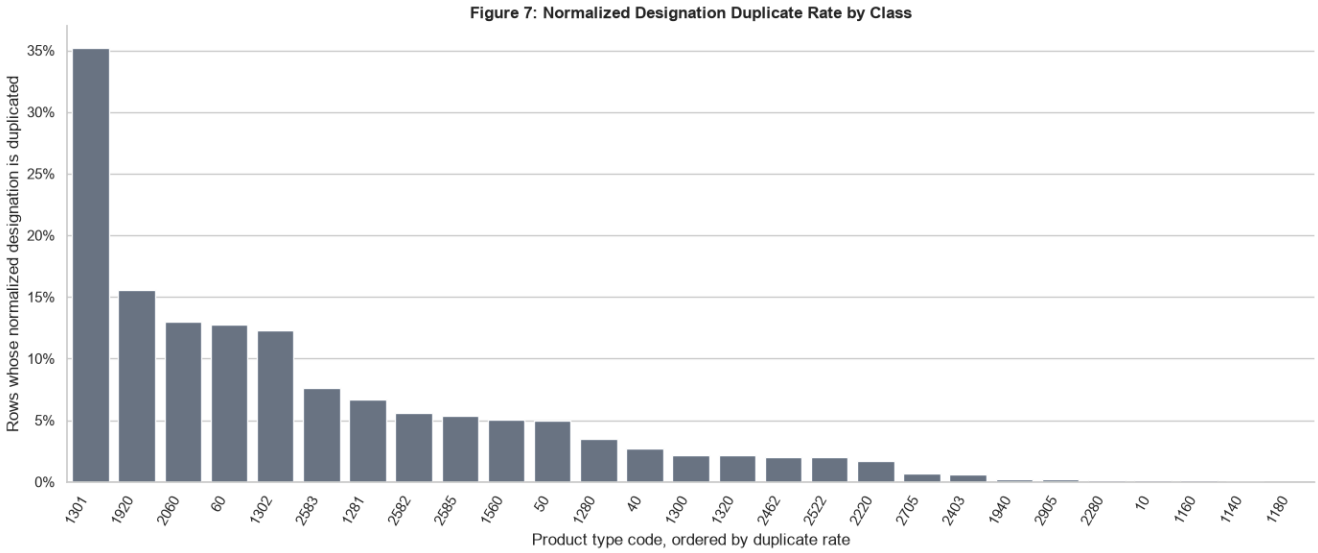
Business implication. Text models should consider both word and character information. For neural models, sequence length should be chosen from the observed distribution.

6. Figure 5 — Relationship Between Designation and Description Length



"Descriptions are much longer than designations, while their lengths are only moderately correlated. Therefore, we should test whether combining both text fields improves classification."

8. Figure 7 — Duplicate Designations and Conflicting Labels



Finding 7 — repeated designations and conflicting labels

Observation. 4,172 training rows (4.9%) belong to a repeated normalized designation. 497 rows belong to duplicate-title groups containing more than one target class.

Quantitative evidence. There are 1,463 repeated-title groups and 160 conflicting-label groups.

Business implication. Review conflicting groups before deciding whether they represent true duplicate records or legitimate catalogue entries.

4.172 Zeilen = 4,9 % des gesamten Trainingsdatensatzes haben einen normalisierten Titel, der mindestens zweimal vorkommt. Von diesen Duplikaten gehören 497 Zeilen zu problematischen Gruppen, bei denen derselbe Titel unterschiedliche Klassenlabels besitzt.

Titel Klasse

"Samsung Galaxy S20" 10 "Samsung Galaxy S20" 10 ← normales Duplikat

"Harry Potter DVD" 40 "Harry Potter DVD" 50 ← conflicting duplicate

A **duplicate group** contains all rows that have the same normalized designation.

Example:

Row	Normalized designation	Class
1	iphone 12 black	10
2	iphone 12 black	10
3	iphone 12 black	10
4	harry potter dvd	40
5	harry potter dvd	50
6	sony headphones	30
7	sony headphones	30

This gives:

- **7 duplicate rows**
- **3 duplicate groups**
  - `iphone 12 black`
  - `harry potter dvd`
  - `sony headphones`

Only one group is conflicting:

- `harry potter dvd` → appears with class 40 and class 50

So:

- Duplicate rows = 7
- Duplicate groups = 3
- Conflicting groups = 1

9. Text-Quality Flags

	train
contains_html	18.42%
contains_url	0.24%
contains_mojibake	1.08%

	flag	train_rows
0	contains_html	15645
1	contains_url	203
2	contains_mojibake	918

10. Training Image Duplicate Audit — File and Pixel Hashes

This section checks **exact image duplicates in the training catalog only**. Two definitions are used:

- **File duplicates:** identical SHA-256 hashes of the complete file bytes.
- **Pixel duplicates:** identical SHA-256 hashes of the decoded RGB dimensions and pixel values.

No perceptual hash is used. The hashes are cached under `data/processed/`, and the raw images are not modified.

```
Images reused from cache: 84,916; to scan: 0
Readable images: 84,916/84,916
```

hash_type	duplicate_groups	duplicate_rows
File SHA-256	3264	8956
Pixel SHA-256	3264	8956

	hash_type	hash_prefix	rows
0	file	1a07ac586847	47
1	file	818f6c09f96c	26
2	file	e811419be694	26
3	file	346527e3aa29	22
4	file	8420074fdaf9	21
5	file	2423a6c4aedc	20
6	file	afa2a66a219b	20
7	file	ce682062fdd8	20
8	file	a8f7abb17656	19
9	file	5f5c00e5cc04	18
10	pixel	b3590c6f8cd0	47
11	pixel	7743c1bbd643	26
12	pixel	a046fed161d0	26
13	pixel	a3e6b6a6ebb6	22
14	pixel	a5f5256e3209	21
15	pixel	106bf57e5a83	20
16	pixel	65b2e38ff376	20
17	pixel	c24f01c7e4f7	20
18	pixel	bad73539181b	19
19	pixel	0f15861ee9ca	18

	products_per_group	duplicate_groups	products_in_groups
0	2	2303	4606
1	3	515	1545
2	4	187	748
3	5	80	400
4	6	41	246
5	7	39	273
6	8	35	280
7	9	16	144
8	10	12	120
9	11	4	44
10	12	3	36
11	13	7	91
12	14	3	42
13	15	5	75
14	16	2	32
15	17	1	17
16	18	2	36
17	19	1	19
18	20	3	60
19	21	1	21
20	22	1	22
21	26	2	52
22	47	1	47

	count
pixel_duplicate_groups	3264
conflicting_pixel_groups	479
rows_in_conflicting_groups	1074

Reading the results

`duplicate_groups` counts distinct training-image hashes that occur more than once. `duplicate_rows` counts all training rows belonging to those groups. The preview lists the ten largest groups for each hash type; the complete lists remain available as `file_duplicate_groups` and `pixel_duplicate_groups`.

If the pixel counts exceed the file counts, some images have identical decoded pixels but different file encodings or metadata. Equal counts mean that the detected pixel duplicates are also byte-identical files.

The group-size table shows how many exact pixel-hash groups contain 2, 3, 4 or more products. A **label conflict** means that the same exact pixel hash occurs with more than one `prdtypecode` in the training data. It is a review candidate, not automatically a labelling error.

11. Preprocessing Decisions Supported by Figures 3+

- Keep `has_description` as a candidate feature.
- Compare designation-only, description-only and combined-text models.
- Use word and character representations for text.
- Treat exact file and pixel duplicates as an audit finding; do not remove rows automatically.
- Keep image paths and loading centralized through `src.data_utils`.
- The training-image file and pixel hash cache is stored under `data/processed/`.